

C Pattern Problems

25 solutions with nested loops — verified code and real program output

CORRECT your code already matched the sample **FIXED** your code corrected, style kept **NEW** missing problem, written for you

#	PROBLEM	YOUR FILE	STATUS
1	Solid Square of Stars	pattern2.c	FIXED
2	Right-Angled Star Triangle	pattern3.c	FIXED
3	Inverted Right-Angled Star Triangle	pattern1.c	FIXED
4	Increasing Number Triangle	pattern4.c	CORRECT
5	Decreasing Number Triangle	pattern5.c	CORRECT
6	Repeated Number Triangle	pattern6.c	CORRECT
7	Inverted Repeated Number Triangle	—	NEW
8	Alphabet Triangle	pattern7.c	CORRECT
9	Repeated Alphabet Triangle	pattern8.c	CORRECT
10	Continuous Number Triangle	pattern9.c	FIXED
11	Right-Aligned Star Triangle	pattern10.c	FIXED
12	Right-Aligned Number Triangle	pattern11.c	CORRECT
13	Pyramid of Stars	pattern12.c	FIXED
14	Inverted Pyramid of Stars	pattern13.c	FIXED
15	Number Pyramid	pattern14.c	FIXED
16	Palindrome Number Pyramid	pattern15.c	FIXED
17	Hollow Square	pattern16.c	CORRECT
18	Hollow Right Triangle	—	NEW
19	0-1 Triangle	—	NEW
20	Odd Number Triangle	—	NEW
21	Even Number Triangle	—	NEW
22	Multiplication Pattern	—	NEW
23	Star Diamond	—	NEW
24	Inverted Star Triangle with Spaces	—	NEW
25	Alternating Star and Number Pattern	—	NEW

Note: your file names were out of order (for example pattern2.c is problem 1, pattern1.c is problem 3), so this PDF numbers everything by the problem list. Problem 7 had no file.

1. Solid Square of Stars

FIXED

Stars were printed as "*" – the sample has a space after each star, so it is now "*" .

CODE

```
#include<stdio.h>
int main(){
    int n=5;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

2. Right-Angled Star Triangle

FIXED

Same issue: "*" changed to "*" to match the sample spacing.

CODE

```
#include<stdio.h>
int main(){
    int n=5;
    for(int i=1;i<=n;i++){
        for(int j=1;j<=i;j++){
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
*
* *
* * *
* * * *
* * * * *
```

3. Inverted Right-Angled Star Triangle

FIXED

Loop used a hard-coded 5 and left n unused (now i=n), and "*" changed to "*" .

CODE

```
#include<stdio.h>
int main(){
    int n=5;
    for(int i=n;i>=1;i--){
        for(int j=1;j<=i;j++){
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
* * * * *
* * * * 
* * * * 
* * * * 
* * * * 
* * * * 
* * * * 
* * * * 
* * * * 
* * * * 
```

4. Increasing Number Triangle

CORRECT

CODE

```
#include<stdio.h>
int main(){
    int n=5;
    for(int i=1;i<=n;i++){
        for(int j=1;j<=i;j++){
            printf("%d ",j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

5. Decreasing Number Triangle

CORRECT

CODE

```
#include<stdio.h>
int main(){
    for(int i=5;i>=1;i--){
        for(int j=1;j<=i;j++){
            printf("%d ",j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

6. Repeated Number Triangle

CORRECT

CODE

```
#include<stdio.h>
int main(){
    int n=5;
    for(int i=1;i<=n;i++){
        for(int j=1;j<=i;j++){
            printf("%d ",i);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

7. Inverted Repeated Number Triangle

NEW

This one was missing from your files, written in your style.

CODE

```
#include<stdio.h>
int main(){
    for(int i=5;i>=1;i--){
        for(int j=1;j<=i;j++){
            printf("%d ",i);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

8. Alphabet Triangle

CORRECT

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        char ch='@';
        for(int j=1;j<=i;j++){
            printf("%c ",ch+1);
            ch++;
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
A
A B
A B C
A B C D
A B C D E
```

9. Repeated Alphabet Triangle

CORRECT

CODE

```
#include<stdio.h>
int main()
{
    int n=5;
    char ch='@';
    for(int i=1; i<=n; i++)
    {
        ch=ch+1;
        for(int j=1; j<=i; j++)
        {
            printf("%c ",ch);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
A
B B
C C C
D D D D
E E E E E
```

10. Continuous Number Triangle

FIXED

The last array element was 6 instead of 5, so row 5 printed 6 numbers (11 ... 16).

CODE

```
#include<stdio.h>
int main(){
    int a[]={1,2,3,4,5};
    int p=1;
    for(int i=0;i<5;i++){
        for(int j=1;j<=a[i];j++){
            printf("%d ",p++);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

11. Right-Aligned Star Triangle

FIXED

Printed " *" so every row had an extra leading space; changed to "* " so row 5 starts at the margin.

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<= 5-i;j++){
            printf(" ");
        }
        for(int j=1;j<=i;j++){
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
      *
     * *
    * * *
   * * * *
  * * * * *
```

12. Right-Aligned Number Triangle

CORRECT

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=5-i;j++){
            printf(" ");
        }
        for(int j=1;j<=i;j++){
            printf("%d ",j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5
```

13. Pyramid of Stars

FIXED

Indent loop ran 5-i times, pushing the whole pyramid one column right; now 4-i.

CODE

```
#include<stdio.h>
int main(){
    int a[]={1, 3 , 5 , 7, 9};
    for(int i=0;i<5;i++){
        for(int j=1;j<=4-i;j++){
            printf(" ");
        }
        for(int j=1;j<=a[i];j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
      *
     ***
    *****
   *******
  *********
```

14. Inverted Pyramid of Stars

FIXED

Same off-by-one in the indent loop (5-i, should be 4-i).

CODE

```
#include<stdio.h>
int main(){
    int a[]={1, 3 , 5 , 7, 9};
    for(int i=4;i>=0;i--){
        for(int j=1;j<=4-i;j++){
            printf(" ");
        }
        for(int j=1;j<=a[i];j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
*****
*****
*****
***
*
```

15. Number Pyramid

FIXED

Digits were printed as "%d " with 2-space indent; the sample has contiguous digits (123) with a 1-space step.

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=5-i;j++){
            printf(" ");
        }
        for(int j=1;j<= 2*i-1;j++){
            printf("%d",j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
 123
 12345
1234567
123456789
```

16. Palindrome Number Pyramid

FIXED

Same as problem 15: contiguous digits and 1-space indent step.

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=5-i;j++){
            printf(" ");
        }
        for(int j=1;j<=i;j++){
            printf("%d",j);
        }
        for(int j=i-1;j>=1;j--){
            printf("%d",j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
 121
 12321
1234321
123454321
```

17. Hollow Square

CORRECT

Reads n with `scanf`, so the output below was generated with input 5.

CODE

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n) printf("* ");
            else printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
* * * * *
*           *
*           *
*           *
* * * * *
```

18. Hollow Right Triangle

NEW

CODE

```
#include<stdio.h>
int main(){
    int n=5;
    for(int i=1;i<=n;i++){
        for(int j=1;j<=i;j++){
            if(j==1 || j==i || i==n){
                printf("* ");
            }
            else{
                printf("  ");
            }
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
*
* *
*  *
*   *
* * * * *
```

19. 0-1 Triangle

NEW

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        int v=i%2;
        for(int j=1;j<=i;j++){
            printf("%d ",v);
            v=1-v;
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

20. Odd Number Triangle

[NEW](#)

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=i;j++){
            printf("%d ",2*j-1);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
1 3
1 3 5
1 3 5 7
1 3 5 7 9
```

21. Even Number Triangle

[NEW](#)

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=i;j++){
            printf("%d ",2*j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10
```

22. Multiplication Pattern

[NEW](#)

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=i;j++){
            printf("%d ",i*j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
2 4
3 6 9
4 8 12 16
5 10 15 20 25
```

23. Star Diamond

[NEW](#)

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=5-i;j++){
            printf(" ");
        }
        for(int j=1;j<=2*i-1;j++){
            printf("*");
        }
        printf("\n");
    }
    for(int i=4;i>=1;i--){
        for(int j=1;j<=5-i;j++){
            printf(" ");
        }
        for(int j=1;j<=2*i-1;j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
  *
 ***
*****
*****
*****
*****
***
  *
```

24. Inverted Star Triangle with Spaces

[NEW](#)

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=i-1;j++){
            printf(" ");
        }
        for(int j=1;j<=6-i;j++){
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
* * * * *
 * * * *
  * * *
   * *
    *
```

25. Alternating Star and Number Pattern

[NEW](#)

CODE

```
#include<stdio.h>
int main(){
    for(int i=1;i<=5;i++){
        for(int j=1;j<=i;j++){
            if(j%2==1){
                printf("1 ");
            }
            else{
                printf("* ");
            }
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
1 *
1 * 1
1 * 1 *
1 * 1 * 1
```